Coding & Solution

Date	30 June 2026
Team ID	Not Provided
Project Name	Smart Lender
Maximum Marks	5 Marks

Coding & Solution

The Smart Lender system is developed using Python, Flask, HTML, CSS, and JavaScript to predict loan eligibility. It preprocesses applicant data, applies a trained Random Forest model, and generates instant prediction results. The application provides a simple web interface for entering applicant details and viewing predictions. The solution is lightweight, easy to maintain, and suitable for academic as well as demonstration purposes.

Instructions:

- Install the required Python libraries using the requirements.txt file.
- Run app.py to launch the Flask application on the local server.
- Open the application in a web browser and access the loan prediction page.
- Enter the applicant's details such as income, loan amount, and credit history.
- Submit the form to generate the loan eligibility prediction.
- View the prediction result and verify whether the applicant is eligible for loan approval.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/josephanusha678-anu/Smart-Lender-Loan-Eligibility-Prediction-System-using-Machine-Learning

Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	Flask, scikit-learn, pandas, NumPy, matplotlib
Key Features Implemented	Loan prediction, data preprocessing, Random Forest model, web interface.
Pending / Incomplete Features	Database integration, user authentication, cloud deployment.
Setup / Run Instructions	Install dependencies, run app.py, open http://127.0.0.1:5000/

Code Quality Checklist

S.N	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The project is organized into separate modules for preprocessing, model prediction, and web interface. The application is easy to understand, modify, and extend with additional machine learning models in the future.